Team 06: the MARKETING role

Scenario: marketing may *segment* customers but must not be able to *identify* them.

PII_TYPE	MARKETING sees	Function
NAME	`Kowalski` -> `K*w*l*k*`	every second character
EMAIL	12 hex characters	pseudonym `LEFT(SHA2(LOWER(val)), 12)`: same input, same output, so customers can be counted and joined
PHONE	`+48 600 100 200` -> `+48*********`	**own idea:** keep only the international dialling code (the country); numbers without `+` are fully masked
ADDRESS	`***`	constant
DOB	`1987-06-14` -> `1980-01-01`	**own idea:** decade of birth (generation cohort)
IP	`10.1.2.3` -> `10.1.0.0`	network only
CARD	`****`	constant

The policies live in `snowflake/sql/mp_pii_string.sql` and `mp_pii_date.sql` (a `WHEN CURRENT_ROLE() = 'MARKETING'` branch). Everything below reads the **same rows** as `PII_ADMIN_USER` (raw) and `MARKETING_USER` (masked), in Snowflake and in Spark through Horizon.

Read **20** customers (batch) and **20** orders (streaming) as raw and as MARKETING_USER, in Snowflake and Spark.

Batch: `SILVER.CUSTOMERS_CURRENT` (tags set by dbt, apply_method MANUAL)

engine str	column str	rows compared int64	result str	example str
snowflake	first_name	20	PASS	'Bartek' -> 'B*r*e*'
snowflake	last_name	20	PASS	'Migała' -> 'M*g*ł*'
snowflake	email	20	PASS	'bartek.migaa14@onet.pl' -> '3487860b7fb6'
snowflake	phone	20	PASS	'+48 504 947 014' -> '+48*********'
snowflake	date_of_birth	20	PASS	datetime.date(1960, 2, 12) -> datetime.date(1960, 1, 1)
snowflake	street	20	PASS	'ul. Konarskiego 07/72' -> '***'
snowflake	postal_code	20	PASS	'57-185' -> '***'
spark	first_name	20	PASS	'Bartek' -> 'B*r*e*'
spark	last_name	20	PASS	'Migała' -> 'M*g*ł*'
spark	email	20	PASS	'bartek.migaa14@onet.pl' -> '3487860b7fb6'
spark	phone	20	PASS	'+48 504 947 014' -> '+48*********'
spark	date_of_birth	20	PASS	datetime.date(1960, 2, 12) -> datetime.date(1960, 1, 1)
spark	street	20	PASS	'ul. Konarskiego 07/72' -> '***'
spark	postal_code	20	PASS	'57-185' -> '***'

14 rows, 5 columns

No selection

14 / 14 PASS

Streaming: `SILVER.ORDERS_LATEST` (tags PROPAGATED from bronze)

engine str	column str	rows compared int64	result str	example str
snowflake	card_last4	20	PASS	None -> '****'
snowflake	ip_address	20	PASS	'223.96.177.249' -> '223.96.0.0'
snowflake	shipping_street	20	PASS	'aleja Bema 95' -> '***'
snowflake	shipping_postal_code	20	PASS	'38-514' -> '***'
spark	card_last4	20	PASS	None -> '****'
spark	ip_address	20	PASS	'223.96.177.249' -> '223.96.0.0'
spark	shipping_street	20	PASS	'aleja Bema 95' -> '***'
spark	shipping_postal_code	20	PASS	'38-514' -> '***'

8 rows, 5 columns
No selection

8 / 8 PASS

Which rows does MARKETING see?

`CUSTOMERS_CURRENT` carries the row access policy `RAP_SUPPORT_COUNTRY` (attached by dbt on `country`):

```
CURRENT_ROLE() IN (<all_rows_roles>)
OR EXISTS (SELECT 1 FROM GOVERNANCE.SUPPORT_COUNTRY_MAP m WHERE m.user_name = CURRENT_USER() AND m.country = countr
```

I added `MARKETING` to `all_rows_roles` (`snowflake/governance.tf`), so it sees **all countries**: segmentation needs the whole customer base, and its protection is the masking, not the row filter. `SUPPORT_USER` is not in that list, so it only sees the countries mapped to it in `SUPPORT_COUNTRY_MAP` (PL, Without the change MARKETING would have seen **zero rows** (the policy is deny-by-default), because it is not listed and not in the map.

PII_ADMIN_USER: Snowflake

COUNTRY str	CUSTOMERS int64
DE	694
NL	690
PL	675

3 rows, 2 columns
No selection

MARKETING_USER: Snowflake

COUNTRY str	CUSTOMERS int64
DE	694
NL	690
PL	675

3 rows, 2 columns
No selection

SUPPORT_USER: Snowflake

COUNTRY str	CUSTOMERS int64
NL	690
PL	675

2 rows, 2 columns
No selection

PII_ADMIN_USER: Spark

country str	customers int64
DE	694
NL	690
PL	675

3 rows, 2 columns
No selection

MARKETING_USER: Spark

country str	customers int64
DE	694
NL	690
PL	675

3 rows, 2 columns
No selection

SUPPORT_USER: Spark

country str	customers int64
NL	690
PL	675

2 rows, 2 columns
No selection

DuckDB still refused, SUPPORT unchanged

> **DuckDB is refused as MARKETING_USER, as expected:** `Binder Error: Unable to plan scan for table CUSTOMERS_CURRENT, but table's config disabled non-server-side scan planning`

> **DuckDB is refused as ANALYST_USER, as expected:** `Binder Error: Unable to plan scan for table CUSTOMERS_CURRENT, but table's config disabled non-server-side scan planning`

SUPPORT_USER (unchanged behaviour):

check str	snowflake str	spark str
first_name = X***	PASS	PASS
email = ***@domain	PASS	PASS
phone = ***dddd	PASS	PASS
dob = Jan 1	PASS	PASS
street = ***	PASS	PASS
countries only PL/NL	PASS	PASS
ip = a.b.x.x	PASS	PASS

7 rows, 3 columns

No selection